

GeoFDI: Fault and Slip Detection as Invariance Testing, with Finite-Sample Guarantees for Legged and Wheeled-Legged Robots

Kaiwen Yang
[affiliation TBD]
[email TBD]

[advisor TBD]
[affiliation TBD]

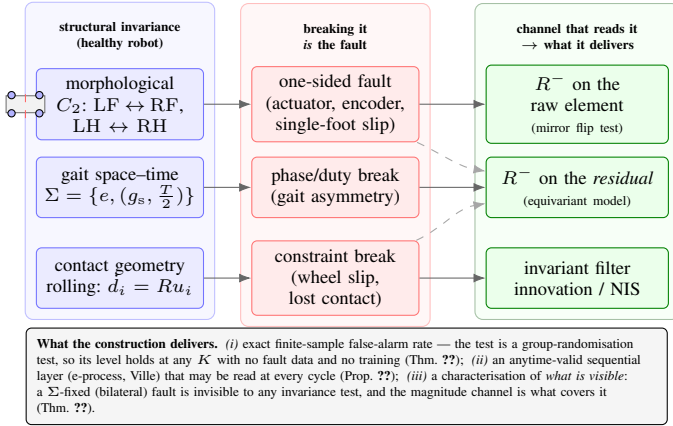


Fig. 1. **Fault and slip detection as invariance testing.** A healthy legged or wheeled-legged robot satisfies three structural invariances (left): the morphological mirror symmetry of the body, the space-time symmetry of the commanded gait, and the geometry of contact (a stance foot is fixed; a rolling wheel’s contact point moves at the measured rolling rate). Every fault or slip we target is a break of one of them (middle), which turns detection into a test of a symmetry hypothesis rather than a comparison against a learned or thresholded nominal. Three channels read the break (right); dashed links mark the couplings that make them complementary rather than redundant.

Abstract—[abstract placeholder]

I. INTRODUCTION

A legged robot’s usefulness outside a laboratory rests on two things it cannot observe directly: whether its actuators still deliver what the controller asks, and whether its feet are where the estimator believes they are. Both degrade silently. An actuator loses torque gradually; a foot slips for a fraction of a step on a wet tile; an encoder drifts by a few milliradians. None of these announce themselves, and each corrupts the state estimate and the controller that depends on it.

The established answers fall into two families, and both leave the same hole. *Model-based* residual methods run a dynamics model — a generalised-momentum observer is the canonical choice [1], [2] — and threshold the residual, usually against a χ^2 quantile. *Data-driven* methods learn what nominal

looks like, from an autoencoder or a recurrent classifier, and threshold a reconstruction error or a predicted fault parameter. The hole is the false-alarm rate. A χ^2 threshold is exact only if the residual is i.i.d. Gaussian, and a legged robot’s residual is neither: it is gait-periodic, serially correlated, and non-stationary across a recording. We measure the consequence directly. On our simulation benchmark the textbook momentum-observer χ^2 detector runs at a nominal alarm rate of **0.15** against a nominal 0.05, and re-calibrating its threshold on the robot’s own nominal data does not repair it — on a *held-out* stretch of the same nominal recording the rate is **0.50**, because the residual’s statistics have already moved (§VI). On an external, third-party fault dataset a Mahalanobis detector fires on **52 %** of healthy segments (0.522 false alarms per healthy gap) where our test fires on **4.9 %**, i.e. at its nominal level (§VII). A detector whose false-alarm rate is unknown cannot be given authority over a controller, which is why slip handling in practice is still a hand-tuned threshold on foot speed [3].

a) Gap: Three literatures each hold one piece and none holds the combination. Morphological-symmetry work in robotics [4] uses a robot’s discrete symmetry group to constrain models and policies, but treats symmetry as an architectural prior, not as a hypothesis to be tested. Fault detection and contact estimation [5], [6] build ever better nominal models and contact classifiers, but calibrate their decisions empirically. Invariance testing in statistics [7], [8] gives exactly the guarantee the robotics side lacks — a group-randomisation test whose level is exact in finite samples, and an anytime-valid sequential form — but has not been connected to a physical invariance a robot actually possesses. The missing step is to notice that a healthy legged robot *already satisfies* a group invariance precisely enough to serve as the null hypothesis, and that the faults we care about are exactly the things that break it.

b) Thesis: We propose to detect faults and slip by testing the structural invariances of the robot itself (Fig. 1). The morphological mirror symmetry of the body, the space-time symmetry of the commanded gait, and the geometry of

contact are properties a healthy machine must satisfy; a one-sided actuator fault, a single-foot slip, or a broken rolling constraint are, by construction, violations of them. Framing detection this way replaces a learned or thresholded nominal with a hypothesis that needs no fault data, no training, and — in its model-free form — no dynamics model, and it inherits an exact finite-sample false-alarm rate from the group-randomisation construction.

c) Contributions:

- 1) **Invariance as the null, with an exact level.** We formulate the healthy-robot hypothesis H_0 as distributional invariance of a phase-registered data element under the gait symmetry group, and test it with a group-randomisation (flip) test whose false-alarm rate is *exact at any sample size*, plus an anytime-valid sequential layer built from e-processes. We also give the recalibrated null H'_0 for the realistic case of a robot that is stably but measurably asymmetric, and a power lower bound with an explicit sample-size rule (§IV; Thm. ??, Prop. ??, Prop. ??).
- 2) **A two-layer characterisation of what is visible, and the architecture it forces.** We prove a dichotomy: a fault fixed by the symmetry group (a bilaterally symmetric degradation) is invisible to *every* invariance test, whatever its magnitude, provided the faulted loop keeps a unique symmetric attractor; anything that breaks the law is detectable by a consistent statistic. The mean-level layer gives the power of the specific paired statistic, and the two layers do not coincide — a zero-mean law change is invisible to the paired statistic and visible to an energy-distance one. This is why the system needs three channels rather than one (§IV; Thm. ??).
- 3) **Equivariant nominal models, and why equivariance is not optional.** When a learned dynamics model is used to form a residual, we show the residual inherits H_0 only if the model is equivariant, quantify the contamination a non-equivariant model injects, and construct an equivariant Deep Lagrangian Network by mirror weight sharing. The effect is not marginal: a plain learned model drives the residual test’s false-alarm rate to 1.00 while its equivariant twin stays at 0.00 (§IV, §VI).
- 4) **Rolling contact preserves the invariant filter, and a five-layer evidence stack.** We prove that a rolling wheel’s moving contact keeps the contact-aided invariant filter group-affine — the invariant error dynamics is unchanged, only the mean and the process noise move — which admits wheeled-legged robots to the same estimator and the same gating front-end as point-foot robots. The claims are evaluated at five levels of increasing externality, including *two* of our own hardware platforms (§VIII).

d) What the evaluation shows: On our own Unitree Go2 corpus — eleven sessions, a machine over a year in service, three sites, two dates two months apart — the naive invariance test rejects on **11/11** sessions, i.e. it sees the machine’s real,

stable left–right asymmetry, and the recalibrated level ν_0 reproduces across the two-month gap to within a factor **1.18**, which is smaller than the session-to-session spread within a single day. On the wheeled-legged M1 the rolling-contact filter recovers **0.99–1.00** of the driven path length where a fixed-foot filter recovers **0.03–0.04**. Used as an estimator front-end, the calibrated per-event gate discards **1.9 %** of nominal contact measurements where the standard fixed foot-speed threshold discards **40.6 %** of them on the same real outdoor data. And on the external benchmark the test holds its level at **0.049** against a nominal 0.05.

II. RELATED WORK

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III. PRELIMINARIES

[placeholder]

IV. THEORY

[placeholder]

V. METHOD

[placeholder]

VI. SIMULATION STUDY

[placeholder]

VII. EXTERNAL BENCHMARKS

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VIII. HARDWARE EVALUATION

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IX. LIMITATIONS

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X. CONCLUSION

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APPENDIX A

PROOFS

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